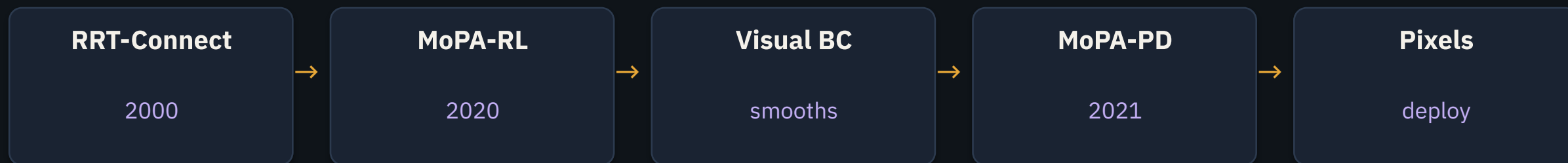


The planner is the teacher, not the product.

MoPA-PD is the CoRL 2021 bridge from Yamada's MoPA-RL to a planner-free visual policy. A contrast pole to teleop VLAs — not a child of JEPA or Cosmos.



Exam bite: MoPA-RL (RRT-Connect + SAC) → visual BC smooths jitter → asymmetric SAC with the BC buffer, weight init, and small α → ~100% on Sawyer clutter, shorter than the expert, DR zero-shot in sim, planner gone at deploy.

Five facts that do not change with the seat

Train privilege

01

Classical MP (RRT-Connect) lives inside MoPA-RL (Yamada et al., CoRL 2020). Geometry and the planner are train-time infrastructure, not the shipped controller.

Distill, then smooth

02

Visual BC on direct-action rollouts removes the planner and smooths RRT jitter. MoPA-PD then runs asymmetric SAC: actor from BC, critic from MoPA-RL Q , $Re:R\pi = 1:3$, small α .

Deploy is planner-free pixels

03

Inference is a 32×32 image plus joints. No RRT-Connect, no object state, no motion planner. The critic's state privilege ends when training ends.

Numbers you must be able to say

04

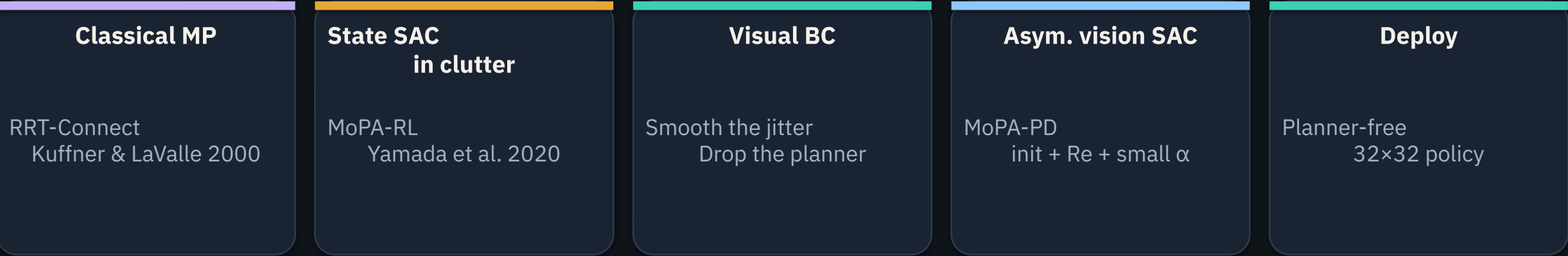
Push 100 / 32.0 / 110.8 (ASR / AEL / ADR). Assembly 0 without BC-smooth vs 100 with. $DR \geq 96.7$ sim-to-sim (Lift Scenario 1 is worst). Expert Push AEL 111 vs Ours 32.

What this paper is not

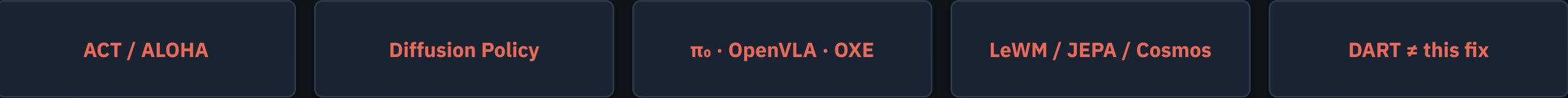
05

Not an ancestor of ACT, Diffusion Policy, π_0 , OpenVLA, or OXE — contrast poles only. Not a JEPA / LeWM / Cosmos child. Sim-only Sawyer. Do not launder DR into sim-to-real.

What descends, in one line



CLEAR CONTRAST — NOT DESCENT



TENTATIVE, and labeled as such: later “privileged sim, vision at deploy” recipes rhyme with this pattern. A specific citation from a 2024–2026 VLA back to MoPA-PD is not established here. Do not draw that arrow.

If it is not in the bibliography, it is not a parent

MoPA-RL, CoRL 2020

Direct parent. MP-augmented SAC. Provides the demos, the critic, and the action-space superset that makes weight init valid.

RRT-Connect, 2000

The classical planner inside the teacher. Collision-free large joint moves. Removed at deploy. Not replaced by a learned planner.

Asymmetric AC, Pinto 2018

Vision actor, state critic. The train/deploy split. Privilege remains during learning, on purpose.

SAC, Haarnoja 2018

Off-policy maximum-entropy actor-critic for both stages. α is a tuned temperature, not a new algorithm.

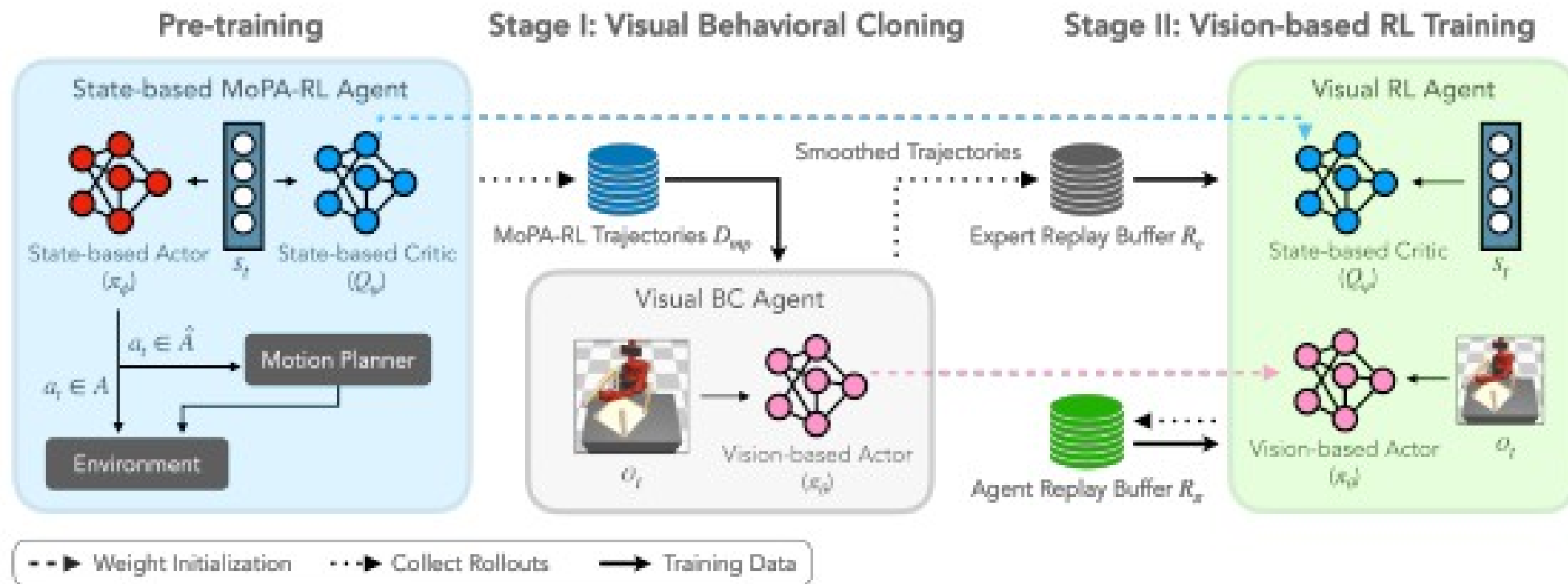
BC / ALVINN, 1989

Pomerleau. Stage 1 is image-action MSE. An old tool, used here as a smoother and an initializer. Not a driving story.

Covariate shift, Ross 2011

Why BC is not enough. The fix is online RL with a smoothed expert buffer — not DAgger queries and not DART noise.

Same picture, read as lineage



Left block

State MoPA-RL. The only place RRT-Connect appears.

Middle block

BC removes the planner and writes R_e . This is the smoothing event.

Right block

The vision actor deploys. The state critic is a train-time wire, not a sensor.

They distill planners. They are not this paper.

What they do

Qureshi et al., Motion Planning Networks, Strudel et al., Jurgenson & Tamar: a network imitates or replaces a planner, often from state or geometry, usually not contact-rich visual manipulation.

What MoPA-PD does

Uses a classical planner only as a teacher inside RL, then throws planning away. The deployed object is a reactive visual policy — not a planner network.

Why the paper bothers

Related work separates those efforts from “distilling motion planners” into visual manipulation policies. Keep the authors’ distinction.

Exam trap

“MoPA-PD is Motion Planning Networks for images.” No. Those models still aim to plan. MoPA-PD aims to stop calling a planner.

DART and DAgger are not the parents of the fix

SHARED

**Ross, Gordon, and Bagnell, 2011:
behavioral cloning fails off the
demonstration manifold.**

MoPA-PD cites that fact, then refuses to stop at BC. Stage 2 rolls the visual policy and mixes a smoothed expert buffer, with no human in the loop.

NOT THE SAME FIX

DAgger

Queries an expert on the learner's states. MoPA-PD does not. The expert is a frozen planner-policy, used offline and as a buffer.

DART

Injects action noise so the expert labels recoveries. Different intervention. Orthogonal sibling, not an ancestor.

CoL

The paper's own IL+RL baseline. It does not solve these tasks. The family is not the solution.

Different data. Different heads. No arrow.

ACT

Action chunking with a CVAE, from human teleoperation. MoPA-PD's actions are single-step joint deltas from a planner-teacher, not chunks from a person.

Diffusion Policy

Denoises an action trajectory conditioned on images, from demonstrations. Table 1's BC-Visual is not this baseline.

π_o / OpenVLA / OXE

Cross-embodiment vision-language-action. One Sawyer, no language, no OXE shard. Absence of a citation is the finding. Do not fill it in.

Allowed sentence

An alternative answer to “where do hard clutter demonstrations come from?” Not a missing chapter in the VLA textbook.

Latent prediction is a different family

LeWM / JEPA

Latent predictive world models. If they plan, they plan in representation space. MoPA-PD's planner is RRT-Connect on joints, and only at train time.

Cosmos

Generative video world models. MoPA-PD does not synthesize futures. It distills a classical planner. Pole, not parent, not child.

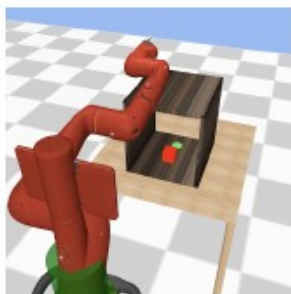
Why the confusion happens

Both say the deployed policy does not call a planner. One amortizes RRT into a CNN. The other never had RRT.

Fence

Do not place MoPA-PD on a JEPA or Cosmos slide except as a labeled contrast pole.

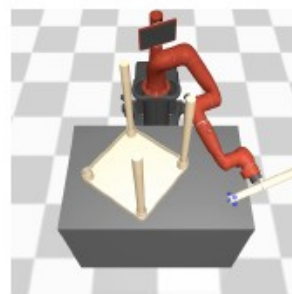
RelMoGen is cited. RoboGen is only a later echo.



(a) Sawyer Push



(b) Sawyer Lift



(c) Sawyer Assembly

RelMoGen · CLEAR cite

Xia et al. 2021. Motion generation inside RL for mobile manipulation. A cousin of the teacher: still wants privilege at deploy.

PBCS · CLEAR cite

Matheron, Perrin, Sigaud. RL and planning for exploration. Same related-work bucket.

RoboGen · parallel

Later systems that keep MP primitives online are a contrast. MoPA-PD's story is that the primitive leaves. No ancestry without a citation.

IKEA env · CLEAR

Assembly sits on Lee, Hu, Lim, ICRA 2021. Task source, not a policy ancestor. Not BEHAVIOR.

The only badges this seat may use

CLEAR Distills MoPA-RL. RRT-Connect at train, gone at deploy.

CLEAR Visual BC smoothing is necessary on Assembly: 0 vs 100.

CLEAR Asymmetric actor-critic, SAC, ALVINN-style BC, Ross covariate shift.

CLEAR DR zero-shot distractors, above 96% ASR, sim-to-sim only.

CLEAR Contrast to ACT, DP, VLAs, JEPA/Cosmos, DART — poles, not parents.

TENTATIVE A soft rhyme with later privileged-sim → vision-deploy recipes.

FALSE Ancestor of π_0 , OpenVLA, JEPA, or Cosmos. Do not say it.

One sentence. Then the fence.

MoPA-PD = MoPA-RL (RRT-Connect + SAC) → visual BC smooths the jitter → asymmetric SAC with the BC buffer, weight init, and a small α → about 100% on Sawyer clutter, shorter than the expert, DR zero-shot in sim, planner gone at deploy.

Same breath: not ACT, not Diffusion Policy, not a VLA ancestor, not a JEPA or Cosmos child. Sim-only. Assembly without smoothing is zero, not “a bit worse.”

High flag. Narrow claim.

Motion planning

RRT-Connect is first-class machinery, and the paper is about removing it. That is the flag. It is not a planner library.

Obstruction

All three tasks are collision-constrained. Baselines without the privileged teacher collapse. That is why this seat exists.

Long horizon, carefully

Lift is reach–grasp–lift. Assembly is an insert among legs. Do not promote this to a TAMP graph or a BEHAVIOR skill.

Leave with

Teacher, smoother, student. Three nouns. No false descendants.